

STA632 Final Term Solved MCQ's

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1. The list of the all units in a population is called.
 - a. Statistic
 - b. Parameter
 - c. Bias
 - d. Sampling Frame
2. Ranked set sampling was introduced by McIntyre in the year
 - a. 1954
 - b. 1950
 - c. 1956
 - d. 1952

alternative to simple random sampling
.. McIntyre (1952) introduced the ba
he population means of pasture and
s
3. Systematic sampling means.
 - a. Selection of n middle units in a sequence.
 - b. Selection of n units situated at equal distance.
 - c. Selection of n largest units.
 - d. Selection of n contiguous units.
4. In random sampling the probability of selecting the item from population is
 - a. Zero
 - b. Known
 - c. Unknown
 - d. One

5. Which of the following would generally require the largest sample size.
- a. Proportional stratified sampling
 - b. Simple random sampling
 - c. Cluster sampling
 - d. Systematic sampling
6. A technique used when selecting cluster of different sizes is called....
- a. Cluster sampling
 - b. Two stage sampling
 - c. One stage sampling
 - d. Probability proportion to size
- 7: Which of the following is not true about stratified random sampling?
- a. It involves a random selection process from identified subgroups
 - b. Proportions of groups in the sample must always match their population proportions
 - c. Disproportional stratified random sampling is especially helpful for getting large enough subgroup samples when subgroup comparisons are to be done
 - d. Proportional stratified random sampling yields a representative sample
8. Which of the following statements are true?
- a. The larger the sample size, the greater the sampling error

- b. The more categories or breakdowns you want to make in your data analysis, the larger the sample needed
 - c. The fewer categories or breakdowns you want to make in your data analysis, the larger the sample needed
 - d. as sample size decreases, so does the size of the confidence interval
9. People who are available, volunteer, or can be easily recruited are used in the sampling method called _____.
- a. Simple random sampling
 - b. Cluster sampling
 - c. Systematic sampling
 - d. Convenience sampling
10. which of the following is the most efficient random sampling technique discussed in your chapter?
- a. Simple random sampling
 - b. Proportional stratified sampling
 - c. Cluster random sampling
 - d. Systematic sampling
11. A number calculated with complete population data and quantifies a characteristic of the population is called which of the following?
- a. A datum
 - b. A statistic
 - c. A parameter
 - d. A population
12. Which of the following would usually require the smallest sample size because of its efficiency?

- a. One stage cluster sampling
 - b. Simple random sampling
 - c. Two stage cluster sampling
 - d. Quota sampling
13. The process of drawing a sample from a population is known as _____.
- a. Sampling
 - b. Census
 - c. Survey research
 - d. None of the above
14. It is recommended to use the whole population rather than a sample when the population size is of what size?
- a. 500 or less
 - b. 100 or less
 - c. 1000 or less
 - d. you should always use a sample
15. Which of the following is not an example of a nonrandom sampling technique?
- a. Purposive
 - b. Quota
 - c. Convenience
 - d. Cluster
16. _____ is a set of elements taken from a larger population according to certain rules.

- a. Sample
 - b. Population
 - c. Statistic
 - d. Element
17. The nonrandom sampling type that involves selecting a convenience sample from a population with a specific set of characteristics for your research study is called ____.
- a. Convenience sampling
 - b. Quota sampling
 - c. Purposive sampling
 - d. Snowball sampling
18. When each member of a population has an equally likely chance of being selected, this is called:
- a. A nonrandom sampling method
 - b. A quota sample
 - c. A snowball sample
 - d. An Equal probability selection method
19. Two stage sampling design is better than the single stage sampling only if the elementary units in the same units are
- a. Positively correlated
 - b. Independent
 - c. Either a or b
 - d. Negatively correlated
20. Double stage sampling is also known as
- a. All of the above

- b. Two directional sampling
- c. Two stage sampling
- d. Two phase sampling

21. What is R command for writing variance-covariance matrix

$$\Sigma = \begin{pmatrix} 1 & 0.95 \\ 0.95 & 1 \end{pmatrix}$$

- a. None of the above
- b. `sig=matrix(c(1,0.95,0.95,1),ncol=4)`
- c. `sig=matrix(c(1,1,0.95,0.95),ncol=2)`
- d. `sig=matrix(c(1,0.95,0.95,1),ncol=2)`

22. Grant estimated the population of England in which year:

- a. In 1952
- b. In 1667
- c. In 1962
- d. In 1662

- Graunt (1662) was the first size of England.

23. Auxiliary variable is also called:

- a. Supporting variable
- b. None of the above
- c. Dependent variable
- d. Sampling variable

24. When Hartley-Ross proposed the unbiased ratio estimator:

- a. In 1964
- b. In 1954
- c. In 1965
- d. None of above

unbiased ratio estimator presented by Hartley and Ross [1954].

25. Hartley-Ross proposed the unbiased regression ratio in which year:

- a. In 1954
- b. In 1965
- c. In 1971
- d. None of the above

26. Estimated value of square root of 800 is:

- a. 2.5495
- b. 28.28
- c. 254.95
- d. 48.48

27. What is the R command to find out variance of x?

- a. Var x
- b. Var(x)
- c. Var()
- d. X=var

28. Simple random sampling and systematic sampling are:

- a. Time sampling
- b. Probability sampling
- c. Quota sampling
- d. Non-probability sampling

29. The estimate of ratio in ratio method of estimation is consistent estimate of population ratio:

- a. True
- b. False

30. What is the R command to find out mean of x?

- a. X=mean
- b. mean(x)
- c. mean x

d. None of the above

31. When every member of the assemble population has an equal chance of being selected to participate in the study, the researcher is using

a. Purposive sampling

b. Simple random sampling

c. Convenience sampling

d. Stratified sampling

32. When the sample size increases, everything else remaining the same, the width of a confidence interval for a population parameter will:

a. Decrease

b. Increase

c. Remain unchanged

d. Sometimes increase and sometimes decrease

33. Mean of cluster mean is the unbiased estimator of population mean when:

a. Clusters are equal sizes

b. Both equal and unequal cluster sizes

c. Clusters are unequal sizes

d. None of the above

34. Which is the R command for selecting a sample of size 10 from the population size

`yp<(111,150,121,198,120,136,140,170,129,115,110,115,121,140)`

a. `s<-sample(yp,1)`

b. `s<-sample(yp,25)`

c. `s<-sample(yp,10)`

d. `s<-sample(yp,5)`

35. If

$$Var(\bar{y}_{sy}) < Var(\bar{y}_{srs})$$

:

a. $\rho\omega=0$

b. $\rho\omega = 1/(nk - 1)$

c. $\rho\omega < -1/(nk - 1)$

d. $\rho\omega > -1/(nk - 1)$

36. We obtain the 10000 random sample of size 6 under SRSWOR using the following population (111,150,121,198,120,136,140,117,186,115,110,115,121,140), which is the R command for repeating this procedure 150 time?

a. `for(i in 1,1500)`

b. `for(i in 1:1500)`

c. `for(i ,1:150)`

d. `for(i in 1:150)`

37. The process of drawing a sample from population is known as.....

a. Survey research

b. Census

c. Sampling

d. None of these

38. A sample is representative part of

a. Population

b. Self-distribution

39. In a sampling with replacement a sampling unit can be selected
- a. Only once
 - b. None of these
 - c. More than one time
 - d. Less than one time
40. In random sampling, the probability of selecting an item from the population is
- a. Un-decided
 - b. Known
 - c. One
 - d. Unknown
41. Bias in which few respondent response to offered questionnaire is classified as:
- a. Distributed error
 - b. Non-responsive bias
 - c. Responsive bias
 - d. Concerning error
42. The cluster 125,115,129,134,111 can be defined in R-Language as:
- a. `Clu<-125,115,129,134,111`
 - b. `Clu<-(125,115,129,134,111)`
 - c. `Clu<-c(125,115,129,134,111)`
 - d. `Clu<-k(125,115,129,134,111)`
43. Which one of the following is not a type of non-probability sampling?
- a. Snowball sampling

- b. Quota sampling
- c. Convenience sampling
- d. Systematic sampling

44. In systematic sampling value of k is classified as:

- a. Multistage interval
- b. Secondary stage interval
- c. Sampling interval
- d. Sub-stage interval

45. The cluster population mean in R-Language is defined as:

- a. $\text{pop.mean} = \sum(y_i) / (N * N)$
- b. $\text{pop.mean} < \sum(y_i) / (N * N)$
- c. $\text{pop.mean} = \sum(y_i) / (N * M)$
- d. $\text{pop.mean} = \sum(y_i) / (N)$

46. Mrs. Trahan samples her class by selecting every third person on her class list. Which type of sampling method is this?

- a. Stratified
- b. Simple
- c. Cluster
- d. Systematic

47. Which of the following would usually require the smallest sample size because of its efficiency?

- a. One stage cluster sampling
- b. Quota sampling
- c. Two stage cluster sampling
- d. Simple random sampling

48. If

$$\sum_{i=1}^4 \sum_{j=1}^5 y_{ij} = 1700, M = 5 \text{ and } n = 4$$

$$\frac{1700}{5 \times 4}$$

Then the weighted mean is:

- a. 85
- b. 80
- c. 82
- d. 90

49. When population is following the linear trend for $n > 1$:

Ans. $Var(\bar{y}_{srs}) < Var(\bar{y}_{sy}) < Var(\bar{y}_{wor})$

50. A sample which is free from bias is called:

- a. Negatively biased
- b. Biased
- c. Positively biased
- d. Unbiased

51. Which ONE of these sampling methods is a probability method?

- a. Judgment sampling
- b. Cluster sampling
- c. Convenience sampling
- d. Quota sampling

52. Another name of standard deviation of a sampling distribution is.....

- a. Standard error
- b. Replication error
- c. Sampling error
- d. Sample standard deviation

53. If $N = 200$, $N_1 = 100$; $n_1 = 15$, the value of w_1 is:

- a. 0.44
- b. 0.21
- c. 0.36
- d. 0.50

$$w_1 = N_1/N;$$

54. What is the value of combined type ratio estimator with

$\bar{Y}_{st} = 10.5$, $\bar{x}_{st} = 50.25$ and $\bar{X} = 45.32$

- a. 9.469
- b. 8.563
- c. 7.258
- d. 10.253

- The combined type ratio estimator is

$$\bar{y}_{rst} = \frac{\bar{y}_{st}}{\bar{x}_{st}} \bar{X}$$

55. Which one of the following methods is generally used in qualitative sampling?

- a. Stratified sampling
- b. Random sampling
- c. Quota
- d. Simple random

56. Farmer Joe separates his farm into 10 regions. He then randomly selected 5 trees from each region to estimate the number of apples produced on his apple tree farm. This issampling.

- a. Systematic
- b. Cluster
- c. Stratified
- d. Simple

57. Mrs. Trahan samples her class by selecting 5 girls and 7 boys. This type of sampling is called?

- a. Stratified
- b. Systematic
- c. Simple
- d. Cluster

58. Mrs. Trahan samples her class by selecting all students sitting at group 1 and group 5 in her classroom. This sampling technique is called?

- a. Simple
- b. Stratified
- c. Systematic
- d. Cluster

59. Which of the following sampling methods is the best way to select a group of people for a study if you are interested in making statements about the larger population?

- a. Convenience sampling
- b. Quota sampling
- c. Purposive sampling
- d. Random sampling.

60. Which of the following statements best describes the relationship between a parameter and a statistic?

- a. A parameter has a sampling distribution with the statistic as its mean.
- b. A parameter has a sampling distribution that can be used to determine what values the statistic is likely to have in repeated samples.
- c. A parameter is used to estimate a statistic.
- d. A statistic is used to estimate a parameter.

61. The value of n_2 by proportional allocation from the following information is: $N_1=580$, $N_2=140$ and $n=80$

$$n_h = \frac{n}{N} N_h$$

a. 19.31

62. In double sampling plan, if the number of defects is in between two cut off numbers c_1 and c_2 then:

Ans. The consumer takes a second sample

63. Among these which sampling is based on equal probability?

a. Systematic sampling

b. Simple random sampling

c. Probability sampling

d. Stratified random sampling

64. The distribution that is formed by all possible values of a statistic is known as:

a. Hypergeometric distribution

b. Sampling distribution

c. Normal distribution

d. Binomial distribution

65. The difference between a statistic and the parameter is called?

a. Probability

b. Sampling error

c. Random

d. Nonrandom

66. The difference between expected value of statistic and the value of the parameter being estimated is called?

- a. Bias
- b. Sampling error
- c. Standard error
- d. Non-random error

67. Increasing the sample size has the following effect upon the sampling error.

- a. It increase the sampling error
- b. All of the above
- c. It reduces the sampling error
- d. It has no effect on sampling error

68. Which of the following is not a type of non-probability sampling:

- a. Stratified random sampling
- b. Snowball sampling
- c. Convenience sampling
- d. Quota sampling

69. Sample is regarded as a subset of?

- a. Data
- b. Set
- c. Distribution
- d. Population

70. Sampling error can be reduced by:

- a. Non-probability sampling
- b. Increasing sampling size
- c. Increasing the population size
- d. Decreasing the sample size

71. Cluster is also known as.....sampling?

- a. Non-probability
- b. Judgmental
- c. Area
- d. Controlled

72. Cluster sampling is used:

- a. When population units are in group
- b. To reduce the sample size
- c. When variance is fixed
- d. To reduce the cost

73. Sampling technique which selects every tenth person from a community is called

- a. Systematic sampling
- b. Cluster sampling
- c. Purposive sampling
- d. Judgmental sampling